StudySapiens Complete Chapter Notes

Chapter 6 — Bioenergetics

Complete Chapter Notes · Textbook topics converted into board-style long questions, with important conceptual short questions inside every topic · PECTAA 2025–26, Punjab English Medium

This chapter follows the textbook topics 6.1 to 6.3. Each topic is converted into board-style long questions, and each topic ends with conceptual short questions. Direct definitions are omitted.

Board pattern (Punjab 2026, new syllabus): Chapter 6 carries 2 MCQs, 3 short questions in Q.4, and part (a) of a long question in Q.7. Shorts are 2 marks; longs are 8 marks.

TOPIC 6.1 — PHOTOSYNTHESIS

Textbook topic 6.1 · printed pages 140–152

Q.1 Explain ATP as the energy currency of cells.

Board long question · 8 marks

Cells store energy by making **ATP** (adenosine triphosphate) and release energy by breaking ATP. ATP is a nucleotide with adenine, ribose and three phosphate groups. Fritz Lipmann proposed that ATP is the universal energy currency of cells.

When ATP is hydrolysed, it becomes **ADP** and inorganic phosphate (Pi), releasing about **7.3 kcal/mole** of energy. This energy drives cellular work. ATP is regenerated by combining ADP with Pi using energy from photosynthesis or respiration.

ATP → ADP + Pi + energy → ATP regenerated

ATP–ADP cycle

ATP transfers energy between metabolic reactions: energy-releasing reactions make ATP, and energy-requiring reactions use ATP.

Q.2 Describe the role of light, carbon dioxide and water in photosynthesis. How do photosynthetic pigments and photosystems capture light energy?

Board long question · 8 marks

Photosynthesis can be summarized as: **$6CO_2 + 12H_2O$ + light → $C_6H_{12}O_6 + 6O_2 + 6H_2O$**. Carbon dioxide, water and light are reactants; glucose and oxygen are products.

- **Light:** Provides energy absorbed by chlorophyll and converted into chemical energy stored in C–H bonds of organic compounds. Plants convert only about 1–2% of solar energy into food. The **action spectrum** shows that blue and red light are most effective.
- **Carbon dioxide:** Acts as the source of carbon for sugar formation. It enters leaves through stomata and is reduced using ATP and NADPH.

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- **Water:** Provides hydrogen for reduction of CO_2. Oxygen released during photosynthesis comes from water, as shown by Van Neil's hypothesis and isotopic tracer experiments.

Photosynthetic pigments: Chlorophyll a and b absorb light, while accessory pigments such as carotenoids absorb additional wavelengths and transfer energy to chlorophyll a. Pigments are organized into **photosystems** in the thylakoid membranes, each with an antenna complex and a reaction centre.

Q.3 Explain the light-dependent reactions of photosynthesis. Differentiate between cyclic and non-cyclic photophosphorylation.

Board long question · 8 marks

Light-dependent reactions occur in the thylakoid membranes. They involve absorption of light by photosystems, excitation of electrons, and formation of **ATP** and **NADPH**.

Non-cyclic photophosphorylation

Electrons flow from water through photosystem II and photosystem I to $NADP^+$, forming NADPH. Water is split (photolysis), releasing oxygen. The electron flow also creates a proton gradient used to make ATP by chemiosmosis.

Cyclic photophosphorylation

Electrons from photosystem I return to the electron transport chain instead of reducing $NADP^+$. This generates ATP but no NADPH and no oxygen. It occurs when the Calvin cycle slows down and $NADP^+$ is unavailable.

Chemiosmosis

Energy from electron flow pumps H^+ into the thylakoid space. H^+ diffuses back through **ATP synthase**, driving ATP synthesis. Thus light energy is converted into chemical energy in ATP and NADPH.

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Q.4 Describe the Calvin cycle (light-independent reactions).

Board long question · 8 marks

The Calvin cycle occurs in the **stroma** of chloroplasts. It uses carbon from CO_2, energy from ATP, and reducing power from NADPH to make sugar. It does not directly require light, but depends on products of light reactions.

1. **Carbon fixation:** Rubisco combines CO_2 with RuBP to form phosphoglycerate (PGA).
2. **Reduction:** PGA is reduced to glyceraldehyde 3-phosphate (G3P) using ATP and NADPH.
3. **Regeneration:** Some G3P molecules are used to regenerate RuBP using ATP, so the cycle continues.

For every three CO_2 molecules fixed, one G3P molecule can leave the cycle to contribute to glucose and other organic compounds. The cycle must turn six times

to produce one glucose molecule.

Important Conceptual Short Questions — Topic 6.1

S1. Why is ATP called the universal energy currency?

It stores and transfers energy between metabolic reactions. Energy-releasing reactions make ATP, and energy-requiring reactions use ATP.

S2. How does the action spectrum differ from the absorption spectrum?

The absorption spectrum shows wavelengths absorbed by a pigment. The action spectrum shows the effectiveness of different wavelengths in driving photosynthesis.

S3. Why is water considered the source of oxygen in photosynthesis?

Isotopic tracer experiments showed that oxygen released comes from water, not carbon dioxide. Water is split during photolysis to provide electrons and hydrogen.

S4. How do accessory pigments help photosynthesis?

They absorb wavelengths that chlorophyll a cannot absorb efficiently and transfer the energy to chlorophyll a, broadening the range of light used.

S5. Differentiate between cyclic and non-cyclic photophosphorylation.

Non-cyclic produces ATP, NADPH and oxygen, with electrons flowing from water to $NADP^+$. Cyclic produces only ATP, with electrons returning to photosystem I.

S6. Why does cyclic photophosphorylation occur when the Calvin cycle slows down?

$NADP^+$ becomes unavailable because NADPH is not being consumed, so electrons cycle back to generate ATP instead of reducing $NADP^+$.

S7. How does chemiosmosis produce ATP in chloroplasts?

Electron flow pumps H^+ into the thylakoid space. H^+ diffuses back through ATP synthase, driving ATP synthesis.

S8. Why is rubisco important in the Calvin cycle?

It catalyses the fixation of CO_2 by combining it with RuBP, starting the conversion of inorganic carbon into organic sugar.

S9. How are light-dependent and light-independent reactions linked?

Light-dependent reactions produce ATP and NADPH, which are used by the Calvin cycle to reduce CO_2 into sugar.

TOPIC 6.2 — CELLULAR RESPIRATION

Textbook topic 6.2 · printed pages 152–161

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Q.5 Differentiate between aerobic and anaerobic respiration. Describe fermentation.

Board long question · 5 + 3 marks

Feature	Aerobic respiration	Anaerobic respiration
Oxygen	Required	Not required
Breakdown of glucose	Complete oxidation to CO_2 and H_2O	Incomplete breakdown
ATP yield	Net 36 ATP per glucose (textbook value)	Only 2 ATP per glucose
End products	CO_2 and H_2O	Lactic acid or alcohol and CO_2
Site	Cytoplasm and mitochondria	Cytoplasm

Fermentation: In alcoholic fermentation, pyruvic acid is broken down into alcohol and CO_2, as in yeast. In lactic acid fermentation, pyruvic acid is converted into lactic acid, as in muscle cells during extreme physical activity when oxygen supply is insufficient.

Q.6 Describe glycolysis and its importance in cellular respiration.

Board long question · 8 marks

Glycolysis is the breakdown of glucose into two molecules of **pyruvic acid**. It occurs in the **cytosol** and does not require oxygen. It is common to both aerobic and anaerobic respiration.

Preparatory phase

Glucose is phosphorylated using ATP to glucose 6-phosphate, converted to fructose 6-phosphate, phosphorylated again to fructose 1,6-bisphosphate, and split into two three-carbon molecules that become glyceraldehyde 3-phosphate (G3P).

Energy-yielding phase

Each G3P is oxidized, transferring hydrogen to NAD^+ to form NADH. Substrate-level phosphorylation produces ATP. Finally, phosphoenol pyruvic acid (PEP) transfers phosphate to ADP, forming pyruvic acid and ATP.

> **Important**
>
> Net yield of glycolysis: **2 ATP** and **2 NADH** per glucose molecule.

Q.7 Describe pyruvic acid oxidation and the Krebs cycle.

Board long question · 8 marks

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Pyruvic acid oxidation

Pyruvic acid enters the mitochondrion and is converted to **acetyl CoA**. During this process, CO_2 is released and NAD^+ is reduced to NADH. This links glycolysis to the Krebs cycle.

Krebs cycle

Acetyl CoA combines with oxaloacetic acid to form citric acid. Through a series of reactions, citric acid is oxidized, releasing CO_2 and transferring hydrogen to NAD^+ and FAD. One ATP is produced by substrate-level phosphorylation per turn, and oxaloacetic acid is regenerated.

For each glucose molecule, the Krebs cycle turns twice, producing **2 ATP**, **6 NADH** and **2 $FADH_2$**, along with CO_2.

Q.8 Explain the electron transport chain and chemiosmosis. How is the net yield of 36 ATP calculated?

Board long question · 8 marks

NADH and $FADH_2$ pass energy-rich electrons to the **electron transport chain** in the inner mitochondrial membrane. As electrons move through carriers, energy is used to pump H^+ into the intermembrane space. H^+ diffuses back through **ATP synthase**, driving ATP synthesis. Oxygen is the final electron acceptor and forms water.

> **Important**
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> Textbook balance sheet: each NADH from the Krebs cycle yields **3 ATP**; each $FADH_2$ yields **2 ATP**. Glycolysis NADH yields **2 ATP** each because transport into the mitochondrion costs energy. The net yield of aerobic respiration is **36 ATP** per glucose, making it 18 times more efficient than anaerobic respiration.

Modern estimates often give about 30–32 ATP, but the textbook value of 36 ATP is used for board answers unless otherwise specified.

Q.9 How do other organic molecules serve as fuel for cellular respiration?

Board long question · 8 marks

Cells can use carbohydrates, lipids and proteins as respiratory fuel. Disaccharides and polysaccharides are broken into monosaccharides that enter glycolysis. Lipids are hydrolysed into glycerol and fatty acids; glycerol enters glycolysis as G3P, while fatty acids are converted to acetyl CoA. Proteins are digested to amino acids; after deamination, their carbon skeletons enter as pyruvic acid, acetyl CoA or Krebs cycle intermediates.

Important Conceptual Short Questions — Topic 6.2

S1. Why is aerobic respiration more efficient than anaerobic respiration?

It completely oxidizes glucose to CO_2 and water, releasing much more energy and producing a net 36 ATP, compared with only 2 ATP in anaerobic respiration.

S2. How does fermentation allow glycolysis to continue in the absence of oxygen?

It regenerates NAD^+ by reducing pyruvic acid to lactic acid or alcohol, allowing glycolysis to keep producing ATP.

S3. Why does glycolysis occur in both aerobic and anaerobic respiration?

It is the universal first step of glucose breakdown and does not require oxygen, producing pyruvic acid, ATP and NADH.

S4. How does substrate-level phosphorylation differ from oxidative phosphorylation?

Substrate-level phosphorylation directly transfers a phosphate group from a substrate to ADP. Oxidative phosphorylation uses energy from electron transport and chemiosmosis to make ATP.

S5. Why is oxygen essential for aerobic respiration?

It acts as the final electron acceptor in the electron transport chain, allowing electrons to flow and ATP to be produced. Without oxygen, the chain stops.

S6. How does the Krebs cycle contribute to ATP production?

It produces NADH and $FADH_2$, which carry electrons to the electron transport chain, and also produces some ATP by substrate-level phosphorylation.

S7. Why do glycolysis NADH molecules yield fewer ATP than Krebs cycle NADH in the textbook balance sheet?

They are produced in the cytoplasm and must be transported into the mitochondrion, which costs energy, so each yields 2 ATP instead of 3.

S8. How do lipids provide more energy than carbohydrates?

They contain many carbon-hydrogen bonds, which release large amounts of energy when oxidized.

S9. Why must amino acids be deaminated before entering respiration?

The amino group must be removed before the carbon skeleton can be converted into pyruvic acid, acetyl CoA or Krebs cycle intermediates.

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TOPIC 6.3 — PHOTORESPIRATION

Textbook topic 6.3 · printed pages 161–164

Q.10 Explain photorespiration, its disadvantages and the adaptations of C4 and CAM plants.

Board long question · 8 marks

Photorespiration is a respiratory activity in green cells in the presence of light that releases CO_2. It needs oxygen and produces CO_2 and water, but **no ATP** is produced.

When CO_2 concentration decreases and O_2 concentration increases, **rubisco** adds O_2 to RuBP instead of CO_2. This produces one phosphoglycerate and one phosphoglycolate. Phosphoglycolate is converted through chloroplast, peroxisome and mitochondrion into glycerate, which re-enters the Calvin cycle. In the process, fixed carbon is lost as CO_2.

Disadvantages

C3 plants lose 25–50% of fixed carbon through photorespiration, reducing yields. At higher temperatures, rubisco's oxidative activity increases, making the problem severe in tropical climates.

Adaptations

- **C4 photosynthesis:** Plants such as corn and sugarcane fix CO_2 into a four-carbon compound in mesophyll cells, then release CO_2 in bundle-sheath cells where the Calvin cycle occurs. This concentrates CO_2 and reduces photorespiration.
- **CAM metabolism:** Succulent plants such as cacti open stomata at night and close them during the day. CO_2 is stored as organic acids at night and released for the Calvin cycle during the day, reducing water loss and photorespiration.

Important Conceptual Short Questions — Topic 6.3

S1. Why does photorespiration reduce photosynthetic efficiency?

Rubisco fixes oxygen instead of carbon dioxide, causing loss of fixed carbon as CO_2 without producing ATP or sugar.

S2. How does high temperature increase photorespiration?

At higher temperatures, rubisco's oxidative activity increases relative to its carbon-fixing activity, so more oxygen is fixed instead of CO_2.

S3. How do C4 plants minimize photorespiration?

They concentrate CO_2 in bundle-sheath cells, where the Calvin cycle occurs, reducing rubisco's oxygenase activity.

S4. Why do CAM plants open stomata at night?

Opening stomata at night reduces water loss in hot climates. CO_2 is stored as organic acids and used for photosynthesis during the day.

S5. Differentiate between C3, C4 and CAM pathways.

C3 plants fix CO_2 directly into the Calvin cycle. C4 plants separate initial CO_2 fixation and the Calvin cycle spatially. CAM plants separate them temporally, fixing CO_2 at night.

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